

Radioastronomy

Fortgeschrittenenpraktikum

HS1 Radioastronomy

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1. Introduction

The experiment HS1-Radioastronomy is there to give an overview for operating a radio telescope and to introduce the prerequisites such as characterising and calibrating the telescope before carrying out the experiments. The main experiment to be conducted is the observation of neutral hydrogen in the Milky Way to determine the rotational velocity in order to derive the mass distribution.

1.1. Radioastronomy

1.2. The 21cm Radiation Line

2. Observation and Evaluation

2.1. The KRT3 telescope

2.2. Weather Conditions

The experiment was carried out on 07.09.2026, starting at around 12:35. The weather was stable throughout the day with partial cloud coverage and temperatures around 19°.

2.3. Calibration

As the receiver receives radiation not only from the source T_S but also from the environment T_h . In addition to those two external sources of radiation, the antenna also picks up on the radiation it's own system emits T_{sys} . This means the measured power of the receiver, represented by the antenna temperature, can be broken down the following way:

$$T_A = T_S + T_h + T_{sys} \quad (1)$$

It is difficult to separate the pure system temperature and the background radiation, so these temperatures are usually combined into an effective system temperature:

$$T'_{sys} = T_{sys} + T_h \quad (2)$$

The effective system temperature now includes all unwanted contributions. This means that to know the contribution of a source, one must know the effective system temperature. Determining the effective system temperature is calibrating the telescope by doing a 'On-Off' measurement through measuring a source with a known flux density and then measuring a piece of empty sky. The KRT3 however, can only measure the sun as a source with high enough flux density but the sun is not a constant source which is why a noise generator needs to be used.

The noise generator injects a signal directly into the antenna horn. The calibration temperature $T_{cal} = 170\text{ K}$ is the expected antenna temperature that is observed additionally when the noise generator is switched on.

When assuming the sun to be a point-like source ?? can be used in the form of:

$$T_b \Omega_s = T_A \Omega_A \quad (3)$$

with T_b being the brightness temperature, Ω_s being the solid angle of the sun, T_A being the antenna temperature and Ω_A being the solid angle of the antenna main lobe.

The goal is measuring the antenna temperature of the sun and converting it into a flux density. For this we need the following identities: the antenna theorem $A_e \Omega_A = \lambda^2$, the Rayleigh-Jeans law $B_\nu = 2kT_b/\lambda^2$, the flux density $S_\nu = B_\nu \Omega_s$ and the aperture efficiency $\eta = A_e/A_g = 0.5$ with A_g being the geometric

antenna surface assumed to be a circle.

$$T_b \Omega_s = T_A \Omega_A \quad (4)$$

$$\Omega_A = \frac{\lambda^2}{A_e}, \Omega_S = \frac{S_\nu}{B_\nu}, B_\nu = \frac{2k_B T_b}{\lambda^2} \quad (5)$$

$$S_\nu = \frac{T_A \cdot 2k_B}{A_e} = f(T_A, A_e) \quad (6)$$

$$A_e = \pi r^2 \cdot \eta = 1,5^2 \pi \cdot 0,5 = \pi 1,125 \text{ m}^2 \quad (7)$$

$$\Rightarrow T_A = \frac{S_\nu A_e}{2k_B} = \frac{S_\nu \pi 1,125 \text{ m}^2}{2k_B} \quad (8)$$

2.3.1. Determination of the system temperature

2.3.2. Background Radiation

2.4. Determination of the angular resolution of the KRT3

The expected half-width is $q = 1.02\lambda/D = \dots$ with the wavelength $\lambda = \dots$ and the antenna diameter $D = 3 \text{ m}$.

2.5. Determination of the Sensitivity of the instrument

To determine the theoretical sensitivity of the instrument, the weakest detectable flux density ΔS_ν for a given system temperature T'_{sys} can be given by the radiometer equation:

$$\Delta S_\nu = \frac{2k}{A_e} \frac{C_s T'_{sys}}{\sqrt{\Delta\nu\tau}} \quad (9)$$

where $C_s \approx 2$ is a dimensionless instrumental constant and τ the integration time. With Gleichung 9 the limiting sensitivity in a single channel with $\Delta\nu = 61 \text{ kHz}$ and an integration time of $\tau = 1 \text{ s}$ can be calculated to $\Delta S_\nu = \dots$

The measurement is carried out with the radio source Vir A with an elevation of $\dots^\circ$ at the time of measurement. The necessary integration time is calculated to be $\tau = \dots$, taking the flux density at $\nu = 1420,4 \text{ MHz}$ to be $S_\nu = 211 \text{ Jy} \pm 11 \text{ Jy}$ after Baars et al. (1977).

2.6. Observation of neutral hydrogen in the Milky Way

The preamplifier has a frequency range of 1398.00-1460.56 MHz divided into 1024 channels. The frequency resolution of a single channel (the bandwidth) is calculated ... As we observe the H1 line in a single channel, the velocity resolution is ... If we assume a radial velocity of 0 km/s

3. Results

3.0.1. Determination of the system temperature

3.0.2. Background Radiation

3.1. Determination of the angular resolution of the KRT3

3.2. Determination of the Sensitivity of the instrument

3.3. Observation of neutral hydrogen in the Milky Way

4. Discussion

5. Summary

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Anhang A Tabellen

Tabelle 1: Messdaten des Dipol-Scans zur Bestimmung der kinetischen Energie (Schirm 3).

Dipolstrom I [A]	Vertikale Position y [mm]
2,371	−2,8116
2,278	−4,0231
2,181	−5,4668
2,076	−7,0907
1,963	−8,8247
1,868	−10,0697
1,768	−11,3757
1,678	−12,9589
1,568	−14,3459
1,477	−15,6398
1,372	−17,1793

Anhang B Code